

K-Nearest Neighbor (KNN)

Height (in cms)	Weight (in kgs)	T Shirt Size
158	58	M
158	59	M
158	63	M
160	59	M
160	60	M
163	60	M
163	61	M
160	64	L
163	64	L
165	61	L
165	62	L
165	65	L
168	62	L
168	63	L
168	66	L
170	63	L
170	64	L
170	68	L

Step 1: Calculate Similarity based on distance function

Euclidean :

$$d(x, y) = \sqrt{\sum_{i=1}^m (x_i - y_i)^2}$$

Manhattan / city - block :

$$d(x, y) = \sum_{i=1}^m |x_i - y_i|$$

The idea to use distance measure is to find the distance (similarity) between new sample and training cases and then finds the k-closest customers to new customer in terms of height and weight.

New customer named 'Monica' has height 161cm and weight 61kg.

Euclidean distance between first observation and new observation (monica) is as follows –

$$= \text{SQRT}((161-158)^2 + (61-58)^2)$$

Similarly, we will calculate distance of all the training cases with new case and calculates the rank in terms of distance.

The smallest distance value will be ranked 1 and considered as nearest neighbor.

Step 2 : Find K-Nearest Neighbors

Let k be 5. Then the algorithm searches for the 5 customers closest to Monica, i.e. most like Monica in terms of attributes, and see what categories those 5 customers were in.

If 4 of them had 'Medium T shirt sizes' and 1 had 'Large T shirt size' then your best guess for Monica is 'Medium T shirt'.

		=SQRT((A\$21-A6)^2+(B\$21-B6)^2)				
	A	B	C	D	E	
1	Height (in cms)	Weight (in kgs)	T Shirt Size	Distance		
2	158	58	M	4.2		
3	158	59	M	3.6		
4	158	63	M	3.6		
5	160	59	M	2.2	3	
6	160	60	M	1.4	1	
7	163	60	M	2.2	3	
8	163	61	M	2.0	2	
9	160	64	L	3.2	5	
10	163	64	L	3.6		
11	165	61	L	4.0		
12	165	62	L	4.1		
13	165	65	L	5.7		
14	168	62	L	7.1		
15	168	63	L	7.3		
16	168	66	L	8.6		
17	170	63	L	9.2		
18	170	64	L	9.5		
19	170	68	L	11.4		
20						
21	161	61				